

2.1/3.1（数据完全一样）

>PJZ10148.1 murein biosynthesis integral membrane protein MurJ [Acinetobacter baumannii ATCC 19606 = CIP 70.34 = JCM 6841]

MALWRSTFIVSAMTMSRLVGLVRDVLLNVFGAGKDFDTFVVAFRIPNFFRRLFAEGAFSQAFIPVLTE
YKTGRAHAIEVQILISRFGCLLTVMTLTTFVAMVLAIAIYMYAPGFHNDPEKFDLAVSMFRLTIPYLMF
MSLTAFASSILNSYGSFASPAFSPVLLNVAMIAGAWWLTPYMAEPIKALGWSVVAAGILQLAIQIPELWR
KNLLIPPKVDFKHEGVERILKLMPLALFGVSVTQINLLNTIWASFMQDGSVSWLYSAERMTELPLGLIG
VAIGTVILPSLSARHAEQDQAKFRSMIDWAAKVIVLVGLPASIALFMLSTPIIQALFQRGEFDLRDTQMT
ALALQCMSAGVISFMLIKVFAPGFYAQQDQTKTPVRVGLMSVAANAILNVVFIGFFKLINWHAHEHMALALA
SSGSALVNAGLLYFYLHKRNIFRFGAHWKKLALQYGLANLAMIAALWFGNLWYNGELSQWVRVAEVLVGLC
VVGVIAYLIGLLLTGFRPRDLKH

>CAD6016886.1 murJ [Escherichia coli]

MNLLKSLAAVSSMTMFSRVLGFARDAIVARIFGAGMATDAFFVAFKLPNLLRRIFAEGAFSQAFVPILAE
YKSKQGEDATRVFVSYSGLLTALAVVTVAGMLAAPWVIMVTAPGFADTADKFALTSQLLKITFPYILL
ISLASLVGAILNTWNRFSIPAFAPTLLNISMIGFALFAAPYFNPPVLALAWAVTVGGVLQLVYQPLHLKK
IGMLVLPRIHFHDAGAMRVVKQMGPAILGVSVSQISLIINTIFASFLASGSVSWMYADRLMEFPSGVLG
VALGTILLPSLSKSFASGNHDEYNRLMDWGLRCLFLLALPSAVALGILSGPLTVSLFQYQKFTAFDALMT
QRALIAYSVGLIGLIVVKVLAPGFYSRQDIKTPVKIAIVTLILTQLMNLAFIGPLKHAGLSLSIGLAACL
NASLLYWQLRKQKIFTQPGWMAFLLRLVVAVLVMMSGVLLGMLHIMPEWSLGTMPWRLLRLMAVVLAGIA
AYFAALAVLGFKVKEFARRTV

>QPV55899.1 murein biosynthesis integral membrane protein MurJ [Pseudomonas aeruginosa]

MNLLKSLAAVSSITMSRLVGFVRDTILARIFGAGLATDAFFVAFKLPNLLRRIFAEGAFSQAFVPILAE
YKNQQGEEATRTFIAYVSGLLTLVLALVTALGILAAPWVIWVTAPGFADTPEKFALTDDLRTVTFPYILL
ISLSSLAGAILNTWNRFSVPFVPTLLNVAMIGFALFLTPYFDPPVMVLGWAVLAGLLQLLYQLPHLRK
IGMLVLPRLNLRDSGVWRVMKLMPLAILGVSVSQISLIINTIFASFLAAGSVSWMYADRLMELPSGVLG
VALGTILLPMLAKTYSNKDRHEYSRLLDWGLRCLFLLVPCSLALAILAEPLTVSLFQYQKFTTVDAAMT
QRALVAYSVGLLGHLVKVLAPGFYAQQNIRTPVKIALFTLVSTQLMNLAFIGPLQHAGLALSIGLAACL
NAGLLYWQLRKQRLYLPPGWAKFLVKLVAVLAMSAVLLATMHWLPAWEQGAMLERFLRLGLLVVAGLL
AYFGMLALLGFRLRDRFSRAVL

分别是 513aa, 511aa, 512aa

论文提到网址用不了不稳定，改用 <https://mafft.cbrc.jp/alignment/server/>结果一样的 Clustal W

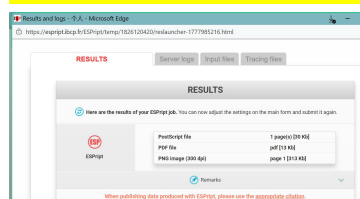


"C:\Users\Lenovo\Downloads_out.2605052133mafftweba3524540P3WRn9rcxHoYy

NOwbsZXklsfnormal.fasta"

符号	含义	生物学意义
*	完全相同	强保守位点（功能关键）
:	强相似	性质接近（可能同功能）
.	弱相似	有点像
空格	不同	变异位点

“用 ESPript3.2 工具（<https://espript.ibcp.fr/ESPrript/cgi-bin/ESPrript.cgi>）可视化”步骤（结果一样，缺参考文献引用）



When publishing data produced with ESPript, please use the appropriate citation.

（提到放图片需要规范引用）

[Citing ESPript - ESPript 3](#)

Citing ESPript

Any published work that makes use of ESPript should cite at least one of the following references:

- Robert, X., Guillon, C. and Gouet, P. (2025) FoldScript: a web server for the efficient analysis of AI-generated 3D protein models. *Nucleic Acids Res.* **53**(W1), W277–W282.
doi: [10.1093/nar/gkaf326](https://doi.org/10.1093/nar/gkaf326) ([freely accessible online](#)).
- Robert, X. and Gouet, P. (2014) "Deciphering key features in protein structures with the new ENDscript server". *Nucl. Acids Res.* **42**(W1), W320–W324.
doi: [10.1093/nar/gku316](https://doi.org/10.1093/nar/gku316) ([freely accessible online](#)).
- Gouet, P., Robert, X. and Courcelle, E. (2003) "ESPrript/ENDscript: extracting and rendering sequence and 3D information from atomic

structures of proteins". *Nucl. Acids Res.* **31**(13), 3320-3323.

doi: 10.1093/nar/gkg556 ([freely accessible online](#)).

- Gouet, P. and Courcelle, E. (2002) "ENDscript: a workflow with web interface to display sequence and structure information". *Bioinformatics* **18**(5), 767-768.

doi: 10.1093/bioinformatics/18.5.767 ([freely accessible online](#)).

- Gouet, P., Courcelle, E., Stuart, D.I. and Metoz, F. (1999) "ESPrript: multiple sequence alignments in PostScript". *Bioinformatics* **15**(4), 305-308.

doi: 10.1093/bioinformatics/15.4.305 ([freely accessible online](#)).

Alignments output layout

Required parameters

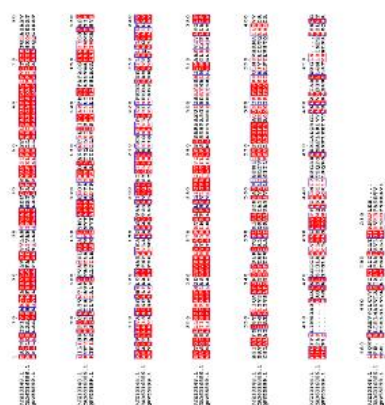
Color scheme ☐ Normal ☐ Flashy ☒ Thermal ☐ Slide ☐ B&W

Orientation ☐ Portrait ☒ Landscape

Paper size ☒ A4 ☐ A3 ☐ A0 ☐ US Letter ☐ Tapestry

Footnotes Suggested number of columns: 75 (for A4, landscape and a font size of 7)

Font size **Number of columns**



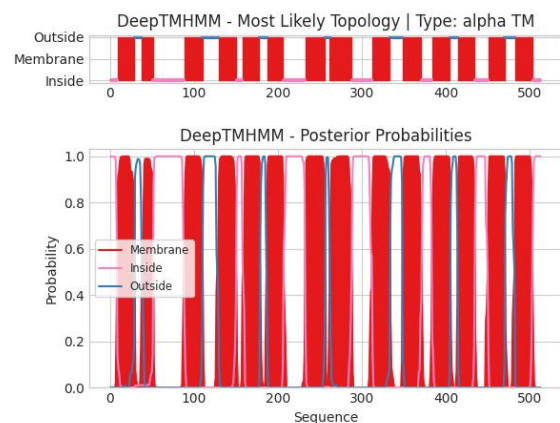
MurJ 在三种菌中高度保守

2.2/3.2（数据完全一样）

利用 ProtParam 对 AbMurJ 蛋白氨基酸序列进行理化性质分析。结果表明，该蛋白由 513 个氨基酸组成，分子量约为 56.84kDa，理论等电点（pI）为 9.53，属于碱性蛋白。不稳定指数为 35.97（<40），表明该蛋白在体外具有良好的稳定性。脂肪族指数为 117.91，显示其具有较高的热稳定性。GRAVY 值为 0.697，表明该蛋白整体呈显著疏水性特征，符合其作为跨膜蛋白的结构属性。氨基酸组成分析显示，Leu（14.0%）、Val（8.6%）和 Ile（6.8%）等疏水性氨基酸含量较高，提示该蛋白富含跨膜 α 螺旋结构单元。此外，正电荷氨基酸数量（39）多于负电荷氨基酸（28），进一步说明其整体呈碱性特征。综上，AbMurJ 蛋白具有典型膜蛋白的理化性质，为后续结构预测及分子对接研究提供了重要理论依据。

2.3/3.3（数据完全一样）

MALWRSTFIVSAMTMLSRVLGLVRDVVLLNVFGAGKDFDTFVVAFRIPNFFR
RLFAEGAFSQAFIPVLTEYKTGRAHAEVQILISRVFGCLLTVMTLTTFVAMVLA
PAIIYMYAPGFHNDPEKFDLAVSMFRLTIPYLMFMSLTAFASSILNSYGSFASPA
FSPVLLNVAMIAGAWWLTPYMAEPIKALGWSVVAAGILQLAIQIPELWRKNLL
IPPKVDFKHGVERILKLMLPALFGVSVTQINLLLNTIWASFMQDGSVSWLYS
AERMTELPLGLIGVAIGTVILPSLSARHAEQDQAKFRSMIDWAAKVIVLVGLPA
SIALFMLSTPIIQALFQRGEFDLRDTQMTALALQCMSAGVISFMLIKVFAPGFY
AQQDTKTPVRVGLMSVAANAILNVVFIGFFKLINWHAEHMALALASSGSALV
NAGLLYFYLHKRNIFRFGAHWKKLALQYGLANLAMIAALWFGLNWyNGELS
QWVRVAEVVGLCVVGVIAYLIGLLLTGFRPRDLKH

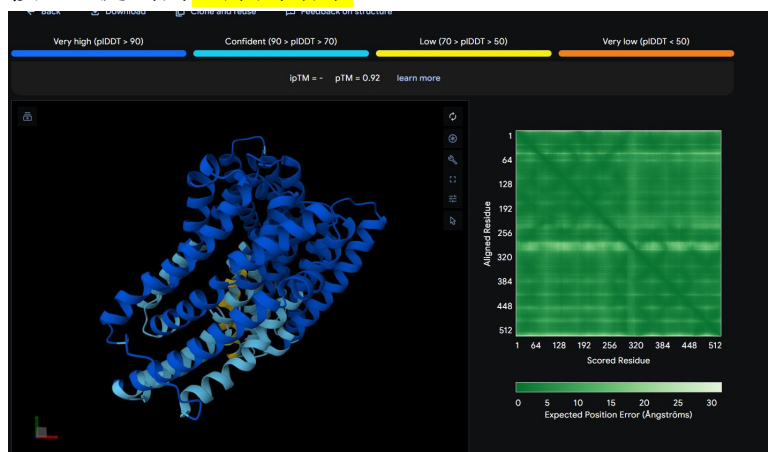


基于 DeepTMHMM 深度学习模型对 AbMurJ 全长氨基酸序列进行跨膜拓扑分析，结果显示该蛋白为典型多跨膜膜蛋白结构，共预测获得 14 个跨膜螺旋（transmembrane helices, TMHs）。其拓扑结构呈现 inside–outside 交替分布模式，N 端与 C 端均定位于胞质侧，符合 MurJ 作为脂质 II 翻转酶的典型膜嵌入特征。该结果表明 AbMurJ 可能通过多跨膜螺旋构建跨膜转运通道，为后续底物识别与功能口袋分析提供结构基础。

输出.csv "C:\Users\Lenovo\Downloads\Unnamed_probs.csv"

AlphaFold Server: 2.1 提到全长氨基酸序列作为本研究的 AbMurJ 蛋白序列，官网找的是 513aa。但是 2.3 这步 AlphaFold Server 依据实验流程 word 并不是这个全序列，而是一个 524aa 的序列。（没有说明这个 524aa 的是怎么来的）可以描述一下：在 N 端起始区域及 C 端柔性尾部存在轻度延伸优化

按 2.1 提到的全序列结果：

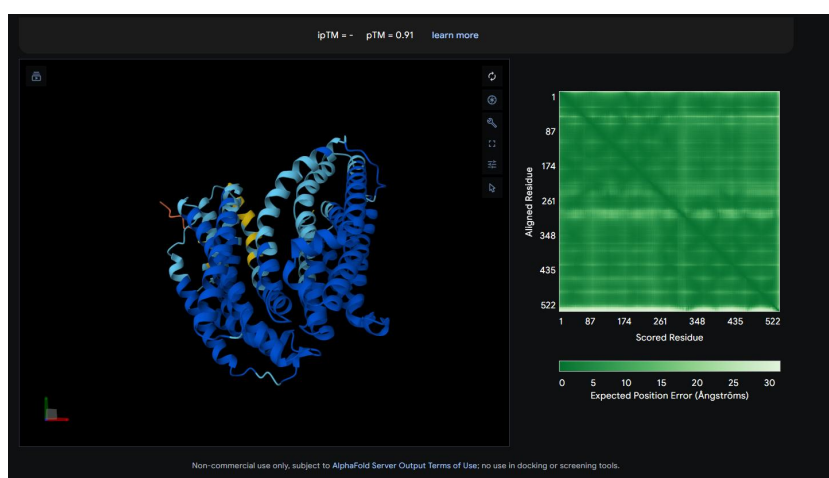


结果 pTM=0.92, 官方序列预测比论文序列 (pTM=0.91) 略高，结构收敛更好

论文序列：

MSTMALWRSTFIVSAMTMLSRVLGLVRDVVLLNVFGAGKDFDTFVVAFRIPN
FFRRLFAEGAFSQAQAFIPVLTEYKTGRAHAEVQILISRVFGCLLTVMITLLTFVAM
VLAPAIYMYAPGFHNDPEKFDLAVSMFRLTIPYLMFMSLTAFASSILNSYGSFA
SPAFSPVLLNVAMIAGAWWLTPYMAEPIKALGWSVVAAGILQLAIQIPELWRK
NLLIPPKVDFKHGVERILKLMLPALFGVSVTQINLLLNTIWASFMQDGSVSW
LYSAERMTELPLGLIGVAIGTVILPSLSARHAEQDQAKFRSMIDWAAKVIVLVG
LPASIALFMLSTPIIQALFQRGEFDLRDTQMTALALQCMSAGVISFMLIKVFAP
GFYAQQDTKTPVRVGLMSVAANAILNVVFIGFFKLINWHAEHMALALASSGS
ALVNAGLLYFYLHKRNIFRFGAHWKKLALQYGLANLAMIAALWFGLNWYN
GELSQQWIRVAEVVGLCVVGVIAYLIGLLLQAGFRPRDLKTTKIKKRSI

结果：（一模一样的）



0.91

Swissmodel 预测：（一模一样的）

1524

Model 01

Structure Assessment

Compare

Download files

Display files

Oligo-State

Monomer

GMQE

0.90

We predict the structure to have a transmembrane segment.

Show / Hide

Template

A0A7U2ZL88.1.A UniProtKB entry unknown, most likely obsolete

AlphaFold DB model of A0A7U2ZL88 (gene: unknown, organism: unknown)

Seq Identity

97.67%

Coverage

Blounit Oligo State

Monomer

QSQE

Method

AlphaFold v2

Seq Similarity

0.60

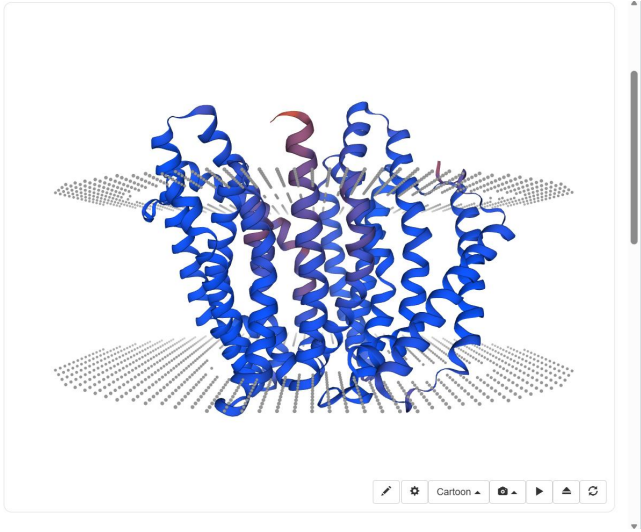
Coverage

0.98

Range

1-515

Model-Template Alignment



Model 02

Structure Assessment

Compare

Download files

Display files

Oligo-State

Monomer

GMQE

0.77

QMEANDisCo Global

0.74 ± 0.05

Membrane annotation has been transferred from template.

Show / Hide

QMEANDisCo Local

QMEAN Z-Scores

Template

7wax.1.A lipid II flippase MurJ

MurJ inward occluded form

Seq Identity

43.37%

Coverage

Blounit Oligo State

Monomer

QSQE

Method

X-ray, 2.35 Å

Seq Similarity

0.42

Coverage

0.96

Range

5-515

Ligand

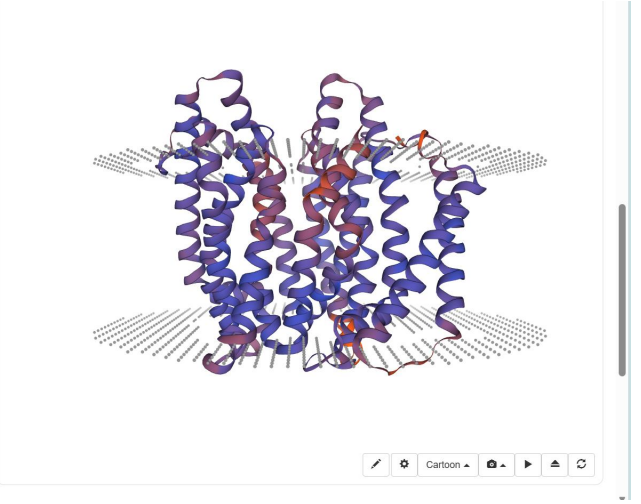
Added to Model

MPD

Not biologically relevant

Description

(4S)-2-METHYL-2,4-PENTANE DIOL



TM-align 对上述两个最优预测模型进行结构比对：

```
*****
*                               *
*   TM-align (Version 20190822)   *
*   An algorithm for protein structure alignment and comparison   *
*   Based on statistics:         *
*   0.0 < TM-score < 0.30, random structural similarity   *
*   0.5 < TM-score < 1.00, in about the same fold   *
*   Reference: Y. Zhang and J. Skolnick, Nucl. Acids. Res. 33, 2302-9 (2005)   *
*   Please email your comments and suggestions to: zhanglab@zhanggroup.org *
*****

Name of Chain_1: A1778070063531440
Name of Chain_2: B1778070063531440
Length of Chain_1: 524 residues
Length of Chain_2: 511 residues

Aligned length= 508, RMSD= 2.91, Seq_ID=n_identical/n_aligned= 0.970
TM-score= 0.8844 (if normalized by length of Chain_1)
TM-score= 0.90549 (if normalized by length of Chain_2)
(You should use TM-score normalized by length of the reference protein)

(":" denotes aligned residue pairs of d < 5.0 Å, "-" denotes other aligned residues)
MSMAL-VRSTFTVYSAMTMSRLVGLQVRYVLLNFGAGQDFDTFYAFRIPNFFRRLFAEGAFSQAFIPVLTEYKGRHAHVQILISRYFGCLLTVMILLTPVAMVLPAITTYMAPGFINDEKFDLAYSMFRLTIPYLMFMSLTAPASSILNYSQSFASFPSPYLLNVAMLAGAW
- A- -LVRSTFTVYSAMTMSRLVGLQVRYVLLNFGAGQDFDTFYAFRIPNFFRRLFAEGAFSQAFIPVLTEYKGRHAHVQILISRYFGCLLTVMILLTPVAMVLPAITTYMAPGFINDEKFDLAYSMFRLTIPYLMFMSLTAPASSILNYSQSFASFPSPYLLNVAMLAGAW
```

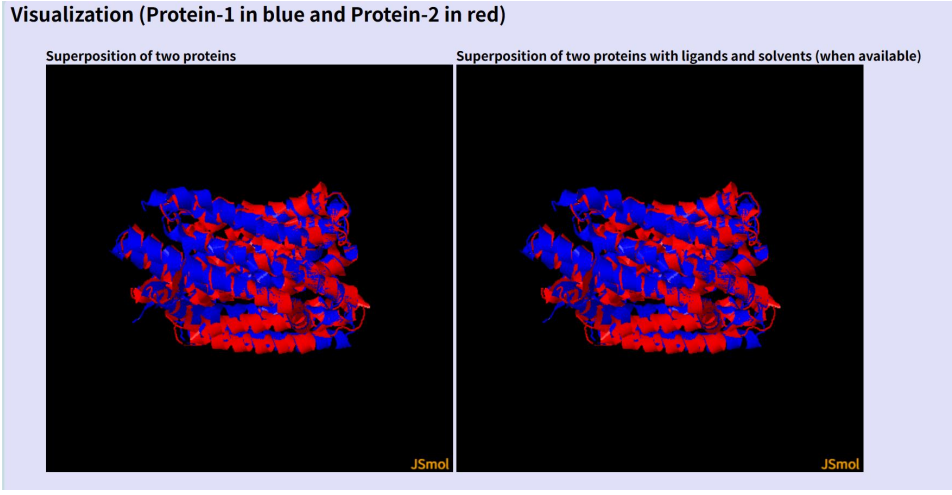
可视化（蛋白质 1 为蓝色，蛋白质 2 为红色）

Superposition of two proteins

两种蛋白质的叠加

Superposition of two proteins with ligands and solvents (when available)

两种蛋白质与配体和溶剂（如有）的叠加



参数	值
Chain_1 长度	524
Chain_2 长度	511
对齐长度	508
RMSD	2.91Å
序列一致性	97.0%
TM-score (以 AF3 模型为参考)	0.884
TM-score (以 SWISS-MODEL 为参考)	0.905

论文结果：

TM-score: 0.906 (以 SWISS-MODEL 模型长度为基准)

RMSD: 2.90 Å

序列一致性: 97.0%

比对长度: 508 个残基

结果和论文基本一致

